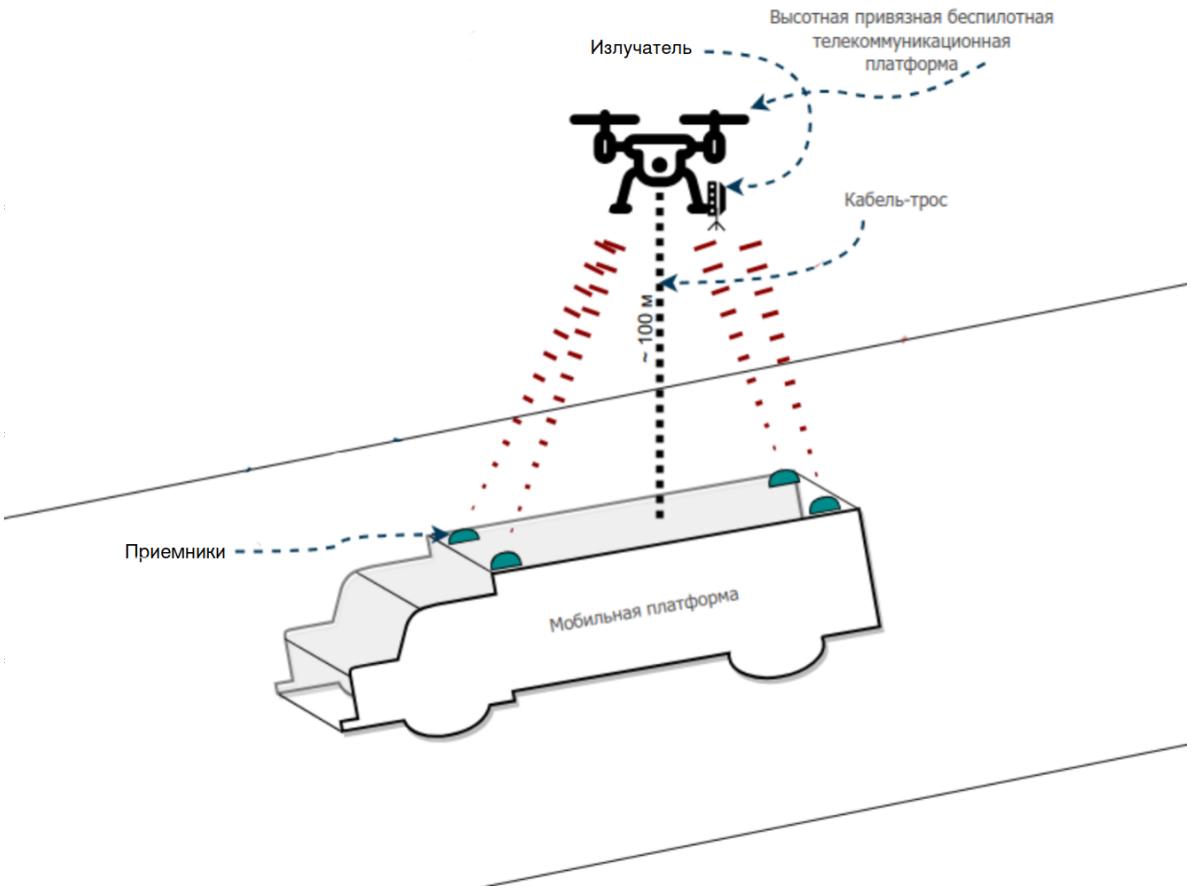
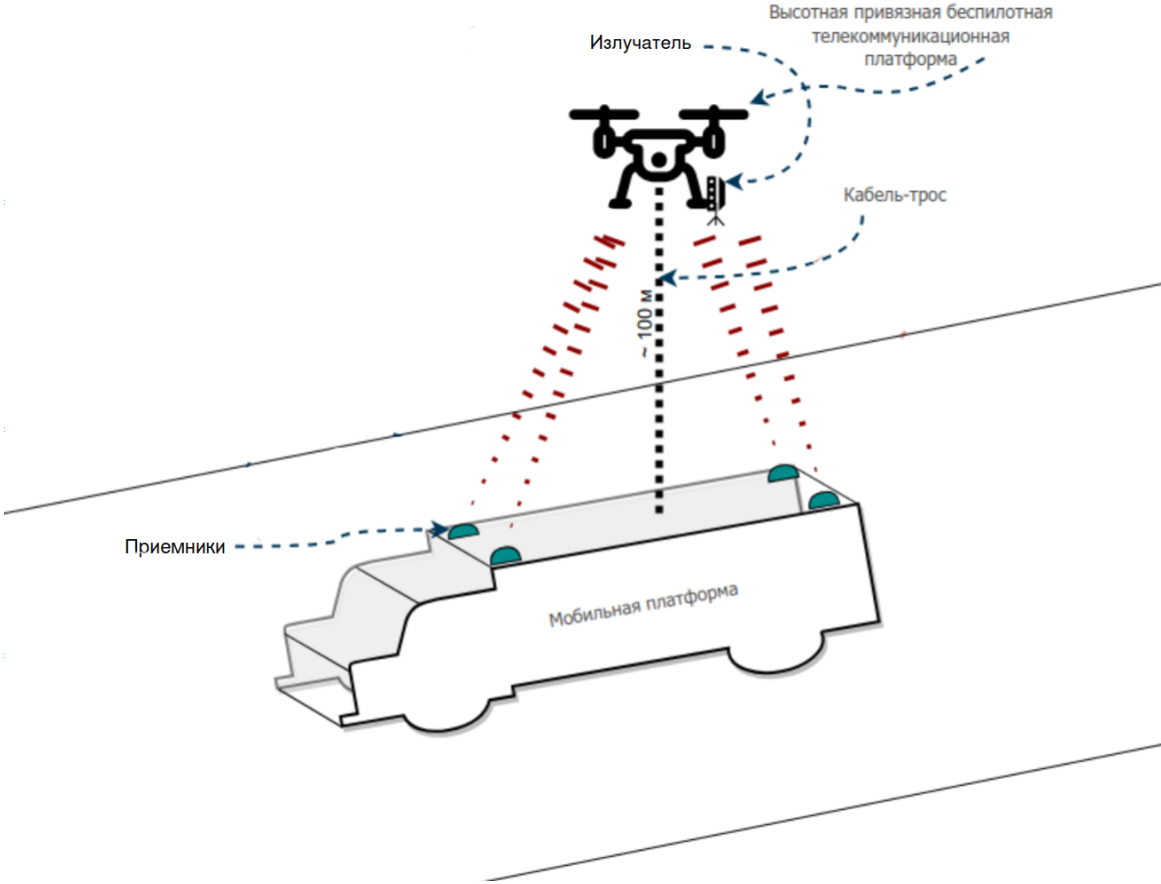
4

, . 69


Высотная привязная беспилотная
телекоммуникационная
платформа
Излучатель
Кабель-трос
~100 м
Приемники
Мобильная платформа

:

- :
  ,
-
- ,


Высотная привязная беспилотная телекоммуникационная платформа
Излучатель
Кабель-трос
~100 м
Приемники
Мобильная платформа

I)

$$I = I_0 e^{-\alpha r},$$

$$\alpha = 8.686\, f^2 \left[\frac{1.84 \times 10^{-11}}{p} \sqrt{\frac{T}{293.15}} \right.$$

$$\left. + \left(\frac{T}{293.15} \right)^{-5/2} \left(\frac{0.01275 e^{-2239.1/T}}{f_{r,\mathrm{O}} + f^2/f_{r,\mathrm{O}}} + \frac{0.1068 e^{-3352/T}}{f_{r,\mathrm{N}} + f^2/f_{r,\mathrm{N}}} \right) \right],$$

- r - , ;
- f — , ;
- T — , ;
- p — , ;
- $f_{r,\mathrm{O}}$, $f_{r,\mathrm{N}}$ — .

:

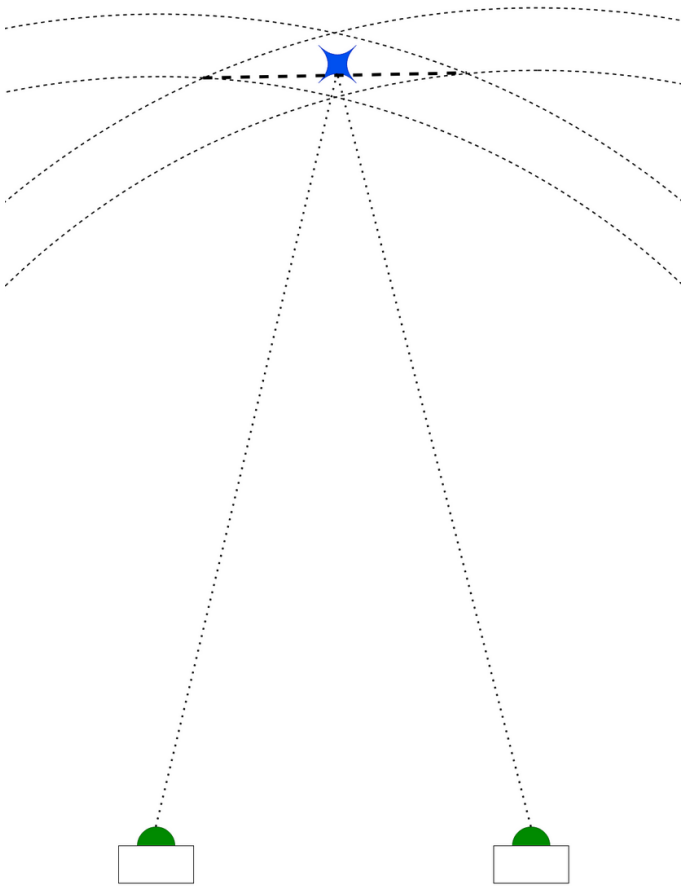
$$\pm d = \frac{c}{2\Delta t} \tag{1}$$

:

Δt	d
1	0.17
10	1.7
100	17
1000	17

, XY

.



-

$$\boxed{\min_{\mathbf{p}} \; J(\mathbf{p}), \qquad J(\mathbf{p}) = \sum_{i=1}^{N} w_i \big(\hat{r}_i - \|\mathbf{p} - \mathbf{a}_i\| \big)^2}$$

- $\hat{r}_i$ — i .
- $\mathbf{a}_i$ — i- .
- $\mathbf{p}$ — .
- w_i — i ( ).
- $J(\mathbf{p})$ — .

P :

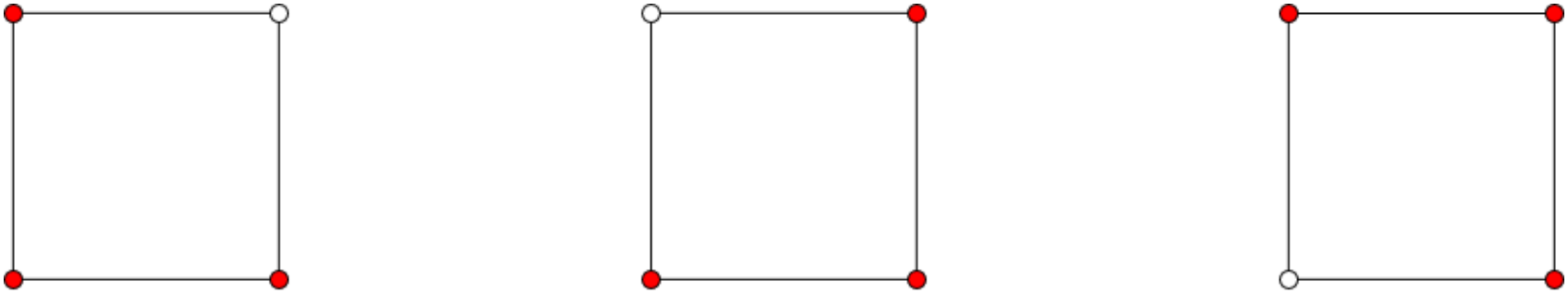
$$(x - x_i)^2 + (y - y_i)^2 + z^2 = r_i^2, \qquad i = 1, 2, 3.$$

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = r_1^2 - r_2^2 + x_2^2 - x_1^2 + y_2^2 - y_1^2,$$
$$2(x_3 - x_1)x + 2(y_3 - y_1)y = r_1^2 - r_3^2 + x_3^2 - x_1^2 + y_3^2 - y_1^2.$$

:

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} = A^{-1} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}.$$

$$z^2 = r_1^2 - (x - x_1)^2 - (y - y_1)^2, z = \pm\sqrt{r_1^2 - (x - x_1)^2 - (y - y_1)^2}.$$



4 Sensors

Statistic	analytic err	gn err
count	2000.0000	2000.0000
mean	0.7494	0.7479
std	0.3575	0.3587
min	0.0346	0.0253
25%	0.4649	0.4627
50%	0.7349	0.7333
75%	1.0091	1.0090
max	1.7822	1.7820

5 Sensors

Statistic	analytic err	gn err
count	2000.0000	2000.0000
mean	0.6755	0.7295
std	0.3292	0.3615
min	0.0399	0.0280
25%	0.4230	0.4511
50%	0.6519	0.7124
75%	0.9082	0.9931
max	1.7357	1.8415

- .
  .
- .
  .
- .
  .
- , . ,
  , .
- . .

!

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As the
was some unprocessed data that should have been added to the final
page this extra page has been added to receive it.
If you rerun the document (without altering it) this surplus page will g
away, because LaTeX now knows how many pages to expect for this
document.